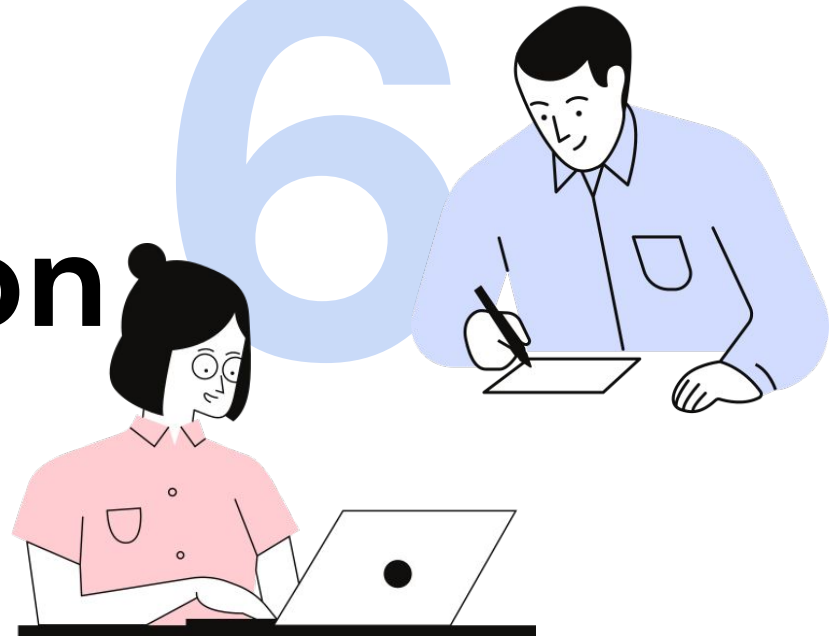




BRITISH INTERNATIONAL
SCHOOL OF SULAYMANIYAH

January
2026

Information & Communication Technology



OVERVIEW

- ❖ This chapter covers a wide range of ICT uses in daily life and industry.
- ❖ Key areas include communication, modelling, and computer control.
- ❖ We will explore financial, medical, and retail applications.
- ❖ The role of satellite and recognition systems is examined.
- ❖ The content connects with previous knowledge on hardware and software.
- ❖ Examples provided are relevant to real-world scenarios.

Modern Communication Methods

- ❖ ICT technology drives various communication systems.
- ❖ Common methods include newsletters, posters, and websites.
- ❖ Multimedia presentations and media streaming are widely used.
- ❖ E-publications are replacing traditional paper formats.
- ❖ Mobile communication has revolutionized how we connect.
- ❖ Each method has specific use cases and target audiences.

Production & Use

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Guidelines for Newsletters

Avoid squeezing too much information onto a single page.

Use clear, easy-to-read fonts like Arial or Trebuchet MS.

Font sizes of 11, 12, or 14 points are recommended.

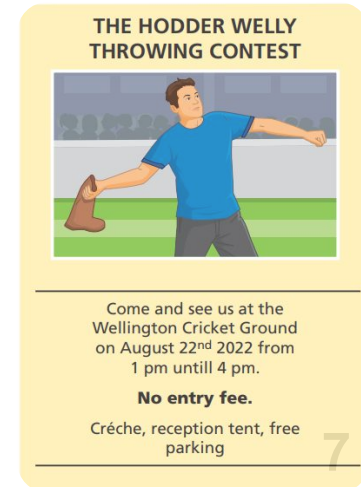
Use columns to organize text and images effectively.

Avoid excessive capital letters; use bold for headings instead.

Real photos are preferred over clip art for engagement.

Essential Information

- ❖ Sporting event posters must include the event type and location.
- ❖ Date, time, and admission fees are critical details.
- ❖ Contact details and facility information (e.g., parking) are needed.
- ❖ Movie posters require a representative image of the genre.
- ❖ Release dates and main character lists are standard.
- ❖ Strategic placement allows targeting of specific audiences.



Websites as opposed to print media

- ❖ Websites offer worldwide advertising capability.
- ❖ No need for paper, consumables, or physical distribution.
- ❖ Companies can develop their own or pay for ad space.
- ❖ Requires investment in design, hardware, and maintenance.
- ❖ Security against hackers and pharming is a necessity.
- ❖ Content management systems make self-hosting easier.

Advantages of Websites

- ❖ Sound, video, and animation can be integrated.
- ❖ Hyperlinks allow easy navigation to related pages.
- ❖ Hit counters provide data on visitor numbers.
- ❖ Global reach extends beyond local markets.
- ❖ Updates are easier and faster than reprinting paper.
- ❖ Content cannot be physically defaced or discarded.

Disadvantages of Websites

- ❖ Risk of hacking, modification, or virus introduction.
- ❖ Pharming attacks can redirect users to fake sites.
- ❖ Customers require internet access and a computer/device.
- ❖ Less portable than paper (though smartphones are changing this).
- ❖ Possible for customers to go to undesirable websites
- ❖ Maintenance costs can be high for professional sites.
- ❖ Harder to target a specific local audience compared to posters.
- ❖ The need to find a way for people to find out about the website

Advantages

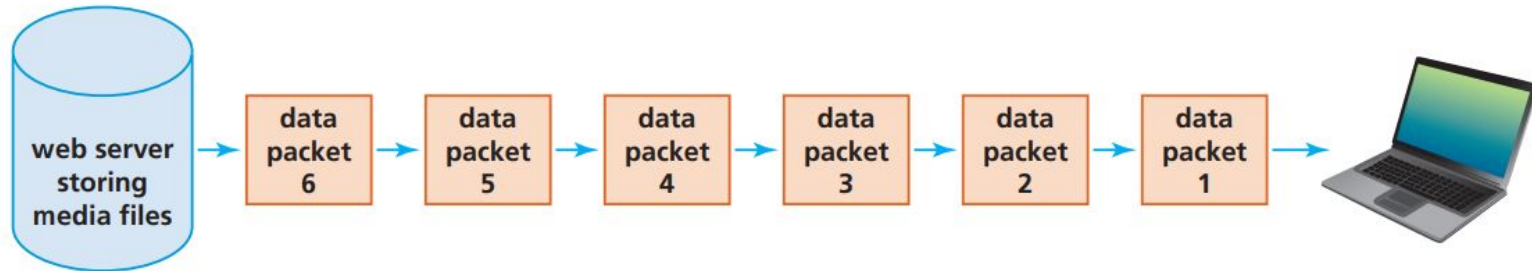
- ❖ Uses animation, video, sound, and music for interest.
- ❖ Produced with software and displayed via projectors.
- ❖ Sound and video grab audience attention effectively.
- ❖ Interactive hyperlinks can access websites or cloud files.
- ❖ Transition effects display facts in chronological order.
- ❖ Presentations can be tailored to different audiences easily.

Disadvantages

- ❖ Specialized equipment can be expensive to procure.
- ❖ Risk of equipment failure during the presentation.
- ❖ Internet access may be required for certain features.
- ❖ Focus can shift to the medium rather than the message.
- ❖ Overuse of effects can result in a "bad" presentation.
- ❖ Too much text or animation can distract the audience.



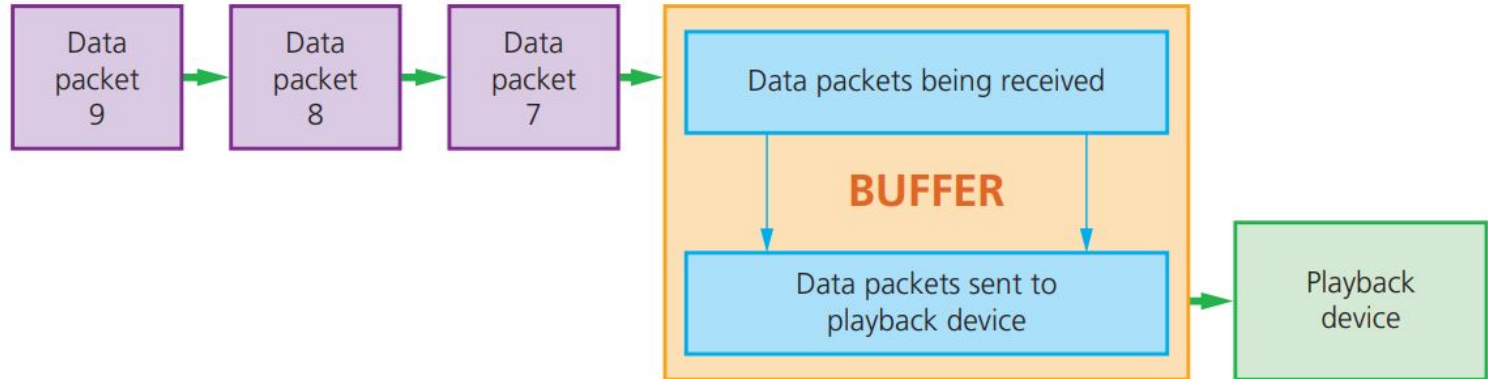
- ❖ Users watch or listen without downloading the full file.
- ❖ Continuous transmission from a remote server to the device.
- ❖ Data is transmitted and played in real-time.
- ❖ Saves local storage space on HDD or SSD.
- ❖ Avoids long wait times associated with full downloads.
- ❖ Requires a stable internet connection (e.g., 25 Mbps for HD).



How Streaming Works

Buffering and Playback

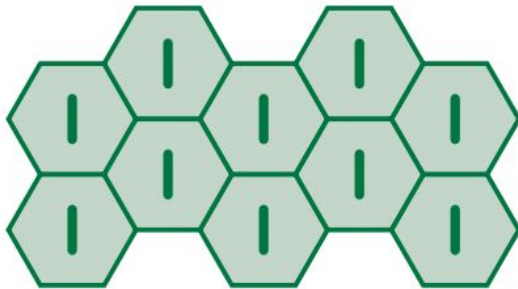
- ❖ Files are sent as a series of data packets.
- ❖ Data packets are buffered (stored temporarily) in the computer.
- ❖ Buffering ensures smooth playback without freezing.
- ❖ The buffer sends data to the playback device continuously.
- ❖ Large buffers reduce the risk of playback gaps.



- ❖ Includes e-books, digital magazines, and newspapers.
- ❖ Downloaded to devices for reading via internet connection.
- ❖ Navigation is done by swiping the screen.
- ❖ Text size can be expanded for better readability.
- ❖ Multimedia elements can be embedded in the text.
- ❖ Cheaper to produce as there are no printing costs.

Cellular Networks

- ❖ Mobile phones use a network of towers inside cells.
- ❖ Towers allow data transmission across large areas.
- ❖ Signals weaken at cell edges and switch to adjacent towers.
- ❖ Satellite technology links calls between different countries.
- ❖ Devices use SIM cards or wireless internet to connect.
- ❖ Supports SMS, calls, VoIP, and internet access.



Cell showing tower at the centre. Each cell overlaps giving mobile phone coverage

- ❖ Quick communication via typed text on mobile devices.
- ❖ Recipients do not need to be immediately available.
- ❖ Often free between phones on the same network.
- ❖ Predictive texting speeds up input by completing words.
- ❖ Useful for communicating when phone calls are not possible.
- ❖ Can be sent even if the recipient's phone is off.

- ❖ Allows staying in touch while on the move.
- ❖ Useful for emergency situations without public phones.
- ❖ Facilitates business and personal calls anywhere.
- ❖ Mobile networks are less stable than landlines.
- ❖ Signal quality depends on network coverage.
- ❖ Essential for keeping contact with co-workers remotely.

Internet Telephony

- Method for talking to people using the internet.
- Converts sound into digital data packets for transmission.
- Accessed via mobile networks or broadband.
- Calls are often free regardless of global location.
- Issues include sound quality, echo, and weird sounds.
- Security risks include identity theft and malware.

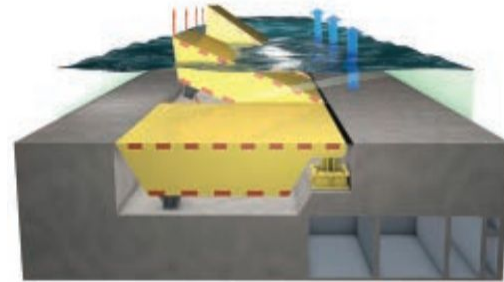
- ❖ Uses built-in cameras to add video to VoIP calls.
- ❖ Cheaper than traditional video conferencing systems.
- ❖ Apps like FaceTime and Zoom are common examples.
- ❖ Split screens allow seeing multiple people simultaneously.
- ❖ Zoom allows recording sessions for later playback.
- ❖ Cloud-based services reduce infrastructure costs.

Spreadsheet Models

- ❖ Spreadsheets calculate profit and loss based on variables.
- ❖ "What-if" scenarios allow testing different financial outcomes.
- ❖ Mathematical formulas represent real-world financial situations.
- ❖ Used to forecast business performance over time.
- ❖ Variables like costs and revenue can be adjusted easily.
- ❖ Essential for budgeting and strategic planning.

- ❖ Engineers use 3D computer modelling for bridges and buildings.
- ❖ Designs are tested long before construction begins.
- ❖ Simulations check for structural integrity under stress.
- ❖ Tests include wind tunnel effects and earthquake resilience.
- ❖ Allows zooming and rotation to view fine details.
- ❖ Identifies design flaws early to save money and lives.

- ❖ Used to predict water levels and potential flood depths.
- ❖ Inputs include river cross-sections and bridge dimensions.
- ❖ Factors like tides and upstream flows are considered.
- ❖ Simulations help plan flood defences and mitigation.
- ❖ Past flooding data calibrates the model for accuracy.
- ❖ Models update continuously as defences are built.



▲ **Figure 6.9** Flood management barriers

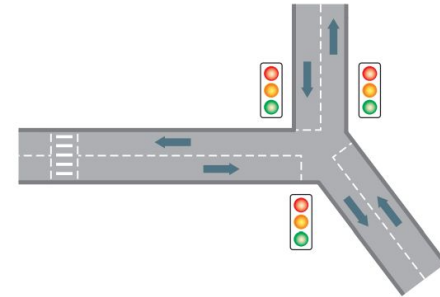
- ❖ Venice uses models to manage flood barriers.
- ❖ Simulations predict barrier reactions to flood scenarios.
- ❖ Helps determine when to raise barriers based on tides.
- ❖ Prevents damage from rising sea levels.
- ❖ Complex calculations are handled by computer systems.
- ❖ Ensures the safety of the city's infrastructure.

Traffic Simulations

- ❖ Models traffic flow during roadworks or accidents.
- ❖ Tests the effect of different speed limits on flow.
- ❖ Helps design road layouts to minimize congestion.
- ❖ Simulates breakdown scenarios to assess impact.
- ❖ Optimization allows for faster repairs and better flow.
- ❖ Cheaper and safer than physical road closures for testing.

Controlling Junctions

- ❖ Models traffic lights at complex junctions (e.g., Y-junctions).
- ❖ Data is collected via induction loop sensors.
- ❖ Records vehicle counts, timing, and build-up at lights.
- ❖ Simulations test light timing variations for efficiency.
- ❖ Considers emergency vehicles and pedestrian crossings.
- ❖ Real-time systems adjust lights based on current traffic.



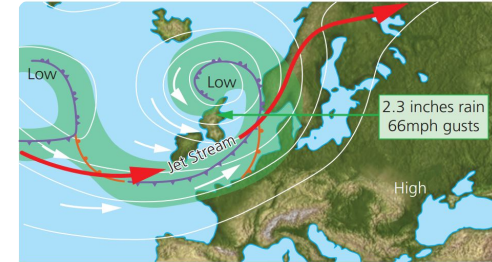
▲ Figure 6.11 Traffic light simulation

- ❖ Counting the number of vehicles passing through the junction from all possible directions.
- ❖ Recording the specific time of day alongside vehicle counts to identify peak hours.
- ❖ Measuring how many vehicles build up (queue) at the junction during different periods.
- ❖ Collecting data across weekends and holidays to account for changing traffic patterns.
- ❖ Timing how long it takes the fastest and slowest vehicles to clear the junction.
- ❖ Accounting for nearby pedestrian crossings, filtering lanes, and specific turn types (left vs. right turns).

- ❖ **Model Validation:** Designers first run the model and compare its results with real-world traffic data to ensure accuracy.
- ❖ **What-if Scenarios:** Once validated, designers vary the timing of the lights to observe the impact on traffic flow.
- ❖ **Dynamic Testing:** Adjusting the number of stopped vehicles or changing traffic volumes in all directions to find the "breaking point."
- ❖ **Emergency Priority:** Simulating how emergency vehicles (ambulances or fire trucks) affect the flow at different times of the day.

- ❖ Weather stations use specialized sensors to automatically gather continuous data on rainfall, temperature, wind, and humidity.
- ❖ Environmental data is gathered every hour of every day to ensure the simulation has a constant stream of real-time information.
- ❖ The atmosphere is divided into a three-dimensional grid where complex mathematical equations represent the exchange of heat and moisture.
- ❖ Forecasters run data through the computer model to simulate air movement and predict weather patterns for several days in advance.

- ❖ The model "learns" by comparing its previous forecasts against actual weather outcomes to improve the accuracy of future predictions.
- ❖ Powerful computers are utilized to perform "number crunching" on vast amounts of data required for complex global simulations.
- ❖ Results are often converted into animated formats, projecting weather symbols onto maps to show real-time changes to the public.



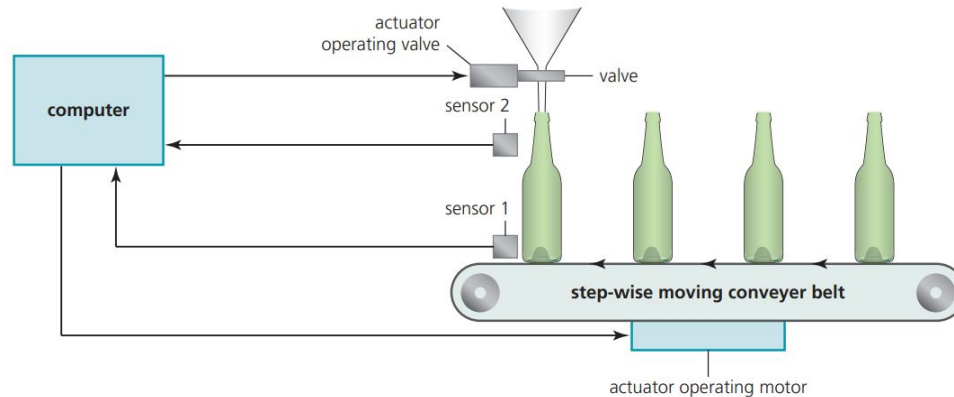
▲ Figure 6.12 Animated weather forecast

Role of Sensors and Actuators

- ❖ Computers use sensors to monitor physical environments.
- ❖ Data is compared against preset values or rules.
- ❖ Decisions trigger actuators to perform tasks.
- ❖ Used in manufacturing, robotics, and automation.
- ❖ Operates continuously without human intervention.
- ❖ Ensures precision and consistency in tasks.



- ❖ Robots handle tasks like filling, capping, and moving bottles.
- ❖ Sensors detect bottle presence and liquid levels.
- ❖ Actuators open/close valves and move conveyor belts.
- ❖ Process is continuous until stopped for maintenance.
- ❖ Ensures consistent filling and production speed.
- ❖ Reduces the need for manual handling of goods.



- ❖ Higher productivity and consistency than human workers.
- ❖ Can work 24/7 without breaks or holidays.
- ❖ Operates safely in hazardous environments (e.g., chemicals).
- ❖ Reduces long-term wage costs for the company.
- ❖ Greater precision minimizes waste and errors.
- ❖ Boredom from repetitive tasks is eliminated.

- ❖ High initial setup and programming costs.
- ❖ Leads to unemployment among manual workers.
- ❖ Robots lack the ability to think or improvise.
- ❖ Maintenance requires expensive, specialized staff.
- ❖ Errors in programming can lead to massive production faults.
- ❖ Security risks if the system is hacked.

- ❖ Tracks daily attendance of students.
- ❖ Methods include OMR, magnetic stripes, and biometrics.
- ❖ Electronic systems are faster than paper calls.
- ❖ Data is stored centrally for easy reporting.
- ❖ Helps identify truancy patterns quickly.
- ❖ Automated messages can be sent to parents.

- ❖ **OMR:** Scans paper registers with pencil marks.
- ❖ **Smart Cards:** Students swipe cards upon entry.
 - Can also be used for cashless catering.
- ❖ **Biometrics:** Fingerprints used for identification.
 - Can prevent students from signing in for others.
- ❖ GPS tracking is a potential (but controversial) future method.

Academic Monitoring

- ❖ Spreadsheets record test results over the year.
- ❖ Teachers compare performance against class averages.
- ❖ Data is easily imported into report cards.
- ❖ Tracks both academic achievement and behaviour.
- ❖ Summaries can be generated for parents or administrators.
- ❖ Identifying underperformance allows for early intervention.

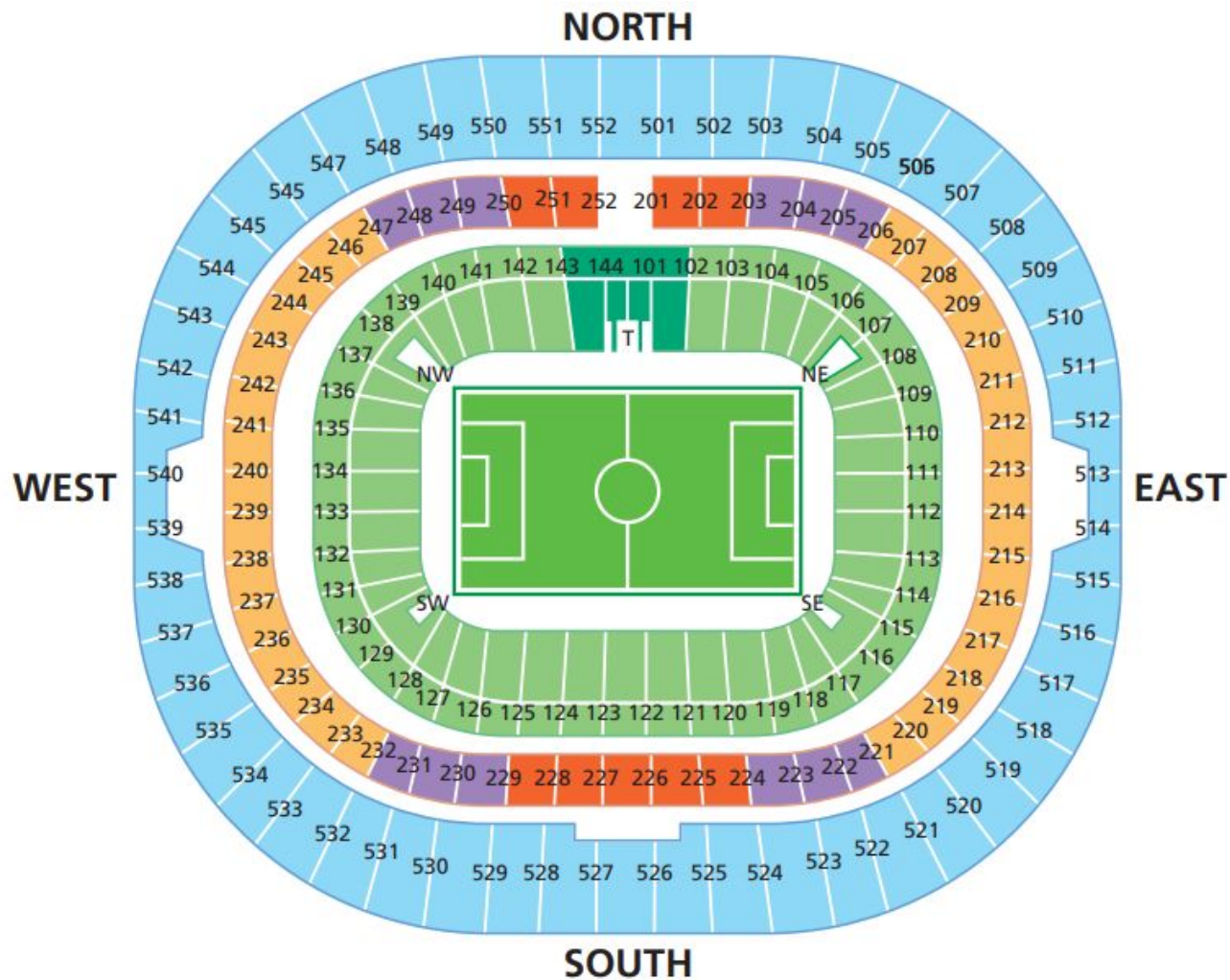
- ❖ Used for booking flights, holidays, and hotels.
- ❖ Customers search for flights by destination and date.
- ❖ Comparison of prices and airlines is simplified.
- ❖ Home printing of boarding passes is possible.
- ❖ Prevents double-booking through real-time databases.
- ❖ Accessible 24/7 from anywhere with internet.

Theatres and Cinemas

- ❖ Users view real-time seating plans.
- ❖ Allows selection of specific seats and prices.
- ❖ E-tickets are sent via email or QR code.
- ❖ QR codes are scanned at the venue entrance.
- ❖ "Impulse" bookings can be made shortly before events.
- ❖ Special offers can be targeted based on booking history.







Advantages of Online Booking

- ❖ Convenience of booking from home at any time.
- ❖ Easy comparison of prices and options.
- ❖ Immediate confirmation and e-ticket delivery.
- ❖ Reduces staffing costs for booking offices.
- ❖ Global audience reach for events and travel.
- ❖ Special offers can be easily distributed.

- ❖ Requires computer/mobile and internet access.
- ❖ Harder to cancel or get refunds compared to person-to-person.
- ❖ Server crashes can prevent bookings entirely.
- ❖ Phishing sites can steal credit card details.
- ❖ Lack of personal interaction for complex queries.
- ❖ System maintenance is expensive.

- ❖ Includes ATMs, online banking, and cheque processing.
- ❖ ATMs allow cash withdrawal and bill payments 24/7.
- ❖ Online banking manages accounts from home.
- ❖ Chip and PIN technology secures card transactions.
- ❖ Electronic funds transfer moves money instantly.
- ❖ Technology has reduced the need for physical branches.

Cheque Processing

- ❖ Cheques are scanned to create a digital image.
- ❖ OCR reads the sort code and account number.
- ❖ Magnetic Ink Character Recognition (MICR) is used for accuracy.
- ❖ "Out clearing" and "in clearing" manage the transfer.
- ❖ Funds are transferred electronically between banks.
- ❖ Digital imaging speeds up the clearing cycle significantly.

Hodder Bank

DATE 1st Feb 2020

PAY H&S Ltd

Fifty dollars only

Account Payee Only

\$ 50.00

SIGNATURE
A.N. Other

11122233344 2020-20 000001

Using Cheques

Advantages	Disadvantages
» more convenient and safer than cash » it is possible to stop payments if necessary » a cheque can be drawn any time (up to six months after it was dated and signed) » cheques can be post-dated » cheques can be traced if they are 'lost'	» cheques are not legal tender and can be refused » it is a slow method of payment » easier for fraudsters than credit card or debit card payment methods » relatively expensive payment method

- ❖ Microchip stores critical account data securely.
- ❖ Users enter a 4-digit PIN to authorize payments.
- ❖ PIN must match the data stored on the chip.
- ❖ More secure than magnetic stripe signatures.
- ❖ Reduces fraud from lost or stolen cards.
- ❖ Readers do not always need a live connection to authorize.

- ❖ Uses Near Field Communication (NFC) technology.
- ❖ Cards or phones are held close to the reader (< 5cm).
- ❖ Transactions are encrypted and dynamic for security.
- ❖ Ideal for small, low-value purchases.
- ❖ Speeds up checkout times significantly.
- ❖ Mobile wallets (e.g., Apple Pay) use this technology.

Type of card	Advantages	Disadvantages
Credit	» there is customer protection if a company stops trading or goods do not arrive » internationally accepted method of payment » interest-free loan if money paid back within agreed time period » can buy items online	» can be charged high interest rate » annual fees often apply » easy to end up with credit damage as sums mount up » security risks when using credit card online (see Chapter 8)
Debit	» money comes from customer's current account, therefore no interest charges » safer than carrying cash » can buy items online	» less customer protection than credit card if goods do not arrive or company goes out of business » no credit allowed; customers must have the funds available » security risks when using debit card online (see Chapter 8)

- ❖ Can withdraw cash at any time of day.
- ❖ Offers banking services without the need to go into the bank.
- ❖ Access an account from anywhere in the world.
- ❖ Provides quicker service than waiting in a queue in a bank.

- ❖ Often in places where theft can take place at night.
- ❖ Potential for shoulder-surfing and card-cloning scams.
- ❖ Banks charge customers for using ATMs.
- ❖ Withdrawal limits are often imposed on customers.
- ❖ If the debit card is faulty then no transaction can take place.
- ❖ Loss of the personal touch (Face to face interactions).